Order types behave as ordinals: equality of order types is isomorphism, membership is representation of an initial segment

The Open Logic Project contributors

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Editorial note. All delivered mathematical text is the author's own, copied from the pinned Open Logic Project unit file sources/olp-set-theory-ordinals/pinned-original-source.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The counted proof is the author's complete argument: the setup with the fixed isomorphism f, the align-style chain of three equivalences, and the closing list of supporting lemmas are all verbatim in the span.

Editorial note. The label ordtypesworklikeyouwant names the counted statement, and the proof environment immediately following it in the source is delivered verbatim as its proof.

Editorial note. The counted statement is carried by the author's own cor environment in the unit source.

Allowed standard prerequisites (editorial).

- Ordinals as representatives of their own isomorphism class. Same unit, \ollabel{ordisoidentity} (source line 599). First line of the proof (line 1075): the identity claim for order types of isomorphic well-orders.
- Uniqueness of isomorphisms between well-orderings. Same unit, \ollabel{ordisounique} (source line 743). Cited at line 1085 for the alignment of restrictions of the fixed isomorphism f to segments.
- Initial segments correspond to ordinal membership. Same unit, \ollabel{wellordinitialsegment} (source line 799). Cited at line 1085: the segment B_{f(alpha)} is handled as the ordinal alpha inside the range of f.
- Ordinals as the set of smaller ordinals. Same unit, \ollabel{ordissetofsmallerord} (source line 433). Cited at line 1085 to read alpha in beta as beta's initial segment at alpha.

Source reference. Original reference ordisoidentity: sources/olp-set-theory-ordinals/pinned-original-source.tex, source lines 599–599. Prerequisite pointer invoked at proof

line(s) 1075; it names a standard fact of the same unit that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original reference ordissetofsmallerord: sources/olp-set-theory-ordinals/pinned-original-source.tex, source lines 433–433. Prerequisite pointer invoked at proof line(s) 1086; it names a standard fact of the same unit that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original reference ordisounique: sources/olp-set-theory-ordinals/pinned-original-source.tex, source lines 743–743. Prerequisite pointer invoked at proof line(s) 1084; it names a standard fact of the same unit that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original reference wellordinitialsegment: sources/olp-set-theory-ordinals/pinned-original-source.tex, source lines 799–799. Prerequisite pointer invoked at proof line(s) 1085; it names a standard fact of the same unit that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Corollary 0.1. *Where* $\langle A, <\rangle$ *and* $\langle B, \lessdot\rangle$ *are well-orderings:*

$$\operatorname{ord}(A,<) = \operatorname{ord}(B,\lessdot) \text{ iff } \langle A,<\rangle = \langle B,\lessdot\rangle$$
$$\operatorname{ord}(A,<) \in \operatorname{ord}(B,\lessdot) \text{ iff } \langle A,<\rangle = \langle B_b,\lessdot_b\rangle \text{ for some } b \in B$$

Proof. The identity holds by ordisoidentity. To prove the second claim, let $\operatorname{ord}(A,<) = \alpha$ and $\operatorname{ord}(B,\lessdot) = \beta$, and let $f\colon \beta \to \langle B,\lessdot\rangle$ be our isomorphism. Then:

$$\begin{aligned} \alpha \in \beta &\text{ iff } f \restriction \alpha\colon \alpha \to B_{f(\alpha)} \text{ is an isomorphism} \\ &\text{iff } \langle A,<\rangle = \langle B_{f(\alpha)}, \lessdot_{f(\alpha)}\rangle \\ &\text{iff } \langle A,<\rangle = \langle B_b,\lessdot_b\rangle \text{ for some } b \in B \end{aligned}$$

by ordisounique, wellordinitialsegment, and ordissetofsmallerord. □